LORENZO SOANNINI

AI, Deep Learning & Computer Vision Engineer

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PROFILE

Computer Science Engineering graduate from Sapienza University of Rome, focused on multimodal learning, generative AI, and computer vision. Developed EEG-to-image reconstruction and 3D medical segmentation pipelines. Winner of Reply Student Tech Clash 2023.

EDUCATION

Master's Degree in Engineering in Computer Science *Sep 2022 – Oct 2026*
Sapienza University of Rome **Grade: 106/110**

- **Thesis:** *EEG-to-Image Reconstruction through Multimodal Visual–Semantic Alignment.* Advisor: Prof. Danilo Comminiello.

Bachelor's Degree in Computer and System Engineering *Sep 2018 – Jul 2022*
Sapienza University of Rome

SELECTED AI & DEEP LEARNING PROJECTS

Cross-Subject EEG-to-Image Reconstruction *(Master's thesis)* Code
PyTorch, LoRA, diffusion models, CLIP

- Developed an EEG-to-image reconstruction pipeline combining a LoRA-adapted **REVE** encoder with diffusion-based image generation on THINGS-EEG2.
- Designed a learned shared/private representation model, improving **CLIP two-way identification from 58.87% to 87.58%** against a residual-decomposition baseline on a matched 100-target evaluation.
- Achieved **87.60% CLIP two-way identification** across 1,654 target images for a held-out participant, using aggregated EEG responses.

3D Liver Tumor Synthesis & Segmentation Code
PyTorch Lightning, MONAI, 3D U-Net

- Adapted the *Label-Free Liver Tumor Segmentation* approach into a modular pipeline for online synthetic lesion generation and **3D U-Net** training on CT scans with liver masks, without manual tumor annotations for training.
- Implemented training and validation loops, sliding-window inference, native-space NIfTI export, and Dice and surface Dice evaluation, with automated checks for lesion generation and spatial alignment.

HACKATHONS & AWARDS

1st Place – Reply Student Tech Clash (Generative AI Edition) *Nov 2023*

- Designed a crisis training simulator for dispatchers in **Italy's 118 emergency medical service**, combining LLMs and expressive text-to-speech to simulate callers experiencing panic, anxiety, and confusion.

Campus Winner & Regional Finalist – Hult Prize (Team Razoo) *2019*

- Won the Sapienza University selection and competed at the European Regional Summit in Oslo, pitching a platform to tackle youth unemployment and streamline SME recruitment.

TECHNICAL SKILLS

- **Frameworks & Libraries:** PyTorch, PyTorch Lightning, MONAI, TensorFlow, OpenCV.
- **Machine Learning:** Multimodal learning, diffusion models (SDXL, unCLIP), LoRA, contrastive learning, representation learning, EEG preprocessing, 3D medical segmentation.
- **Programming:** Python, C, C++, SQL, MATLAB, Java, JavaScript/Node.js.
- **Tools:** Git/GitHub, Linux, ROS, LaTeX.

LANGUAGES & ADDITIONAL INFORMATION

Italian: native. English: C1 listening, reading and writing; B2–C1 speaking. Driving licence: B.